

Compositional Systems in Finite-Dimensional Classical Mechanics

PRINCIPIA PHYSICA research note

16 August 2026

Abstract

We construct a typed mathematical framework for finite-dimensional classical mechanics in which systems are symplectic manifolds, states are points, deterministic processes are smooth maps, observables are smooth real-valued functions, and parallel composition is Cartesian product. We distinguish three claims that are often conflated: products give a symmetric monoidal mathematical structure; additively separable Hamiltonians generate factorized dynamics; and a product represents physically independent subsystems. The first two are proved, whereas the third requires empirical assumptions. Counterexamples show that diagonal constraints, singular constraint sets, and interactions defeat naive product or factorization claims. The result supports compositional description as a precise mathematical organization of classical mechanics, but does not establish that every such structure is empirically realized.

1 Scope and epistemic discipline

This paper concerns smooth, Hausdorff, second-countable, finite-dimensional manifolds without boundary. Field theory, infinite-dimensional mechanics, relativity, and quantum theory are boundary tests rather than members of the formalism. Standard differential-geometric mechanics is cited to Abraham and Marsden [1] and Marsden and Ratiu [4]; categorical terminology follows Mac Lane [3]. Baez and Stay [2] motivate, but do not prove for classical mechanics, the use of symmetric monoidal categories to type systems and processes.

Conjecture 1.1 (Founding thesis under test). *Physics is the study of empirically realized compositional mathematical structures.*

This statement is not used as a premise in any proof below. The paper tests whether its terms can be made precise in a restricted domain.

Assumption 1.2 (Finite-dimensional regularity). *All state spaces and maps used in the core construction are smooth and finite-dimensional; every Hamiltonian vector field under discussion has a flow on the stated time interval.*

The completeness clause is local-in-time when a global flow does not exist.

Axiom 1.3 (Typed equality). *States, processes, observables, and empirical records are distinct types. An equality is admitted only between terms of a common type; a map relating two types is part of the data.*

2 Primitive mathematical data

Definition 2.1 (System and state). *A Hamiltonian-capable system is a pair $S = (M, \omega)$, where M is a finite-dimensional smooth manifold and $\omega \in \Omega^2(M)$ is closed and nondegenerate. A state of S is a point $x \in M$. The unit system is $I = (\{*\}, 0)$.*

Nondegeneracy means that $v \mapsto \iota_v \omega$ is an isomorphism $T_x M \rightarrow T_x^* M$ at every x . The zero form on the zero-dimensional tangent space of I is nondegenerate.

Definition 2.2 (Processes). *For systems $S = (M, \omega)$ and $T = (N, \eta)$, a kinematic process $f : S \rightarrow T$ is a smooth map $f : M \rightarrow N$. A canonical process is a symplectomorphism: a diffeomorphism satisfying $f^* \eta = \omega$. A Hamiltonian time process on S is a map $\Phi_t^H : M \rightarrow M$ in the flow of the vector field X_H defined by*

$$\iota_{X_H} \omega = dH. \quad (1)$$

We use this sign convention throughout.

Kinematic processes permit preparation maps $I \rightarrow S$ and readout maps $S \rightarrow I$, but canonical processes generally do not: positive-dimensional symplectic manifolds are not symplectomorphic to a point. This distinction prevents a single arrow class from silently serving incompatible operational roles.

Definition 2.3 (Observables and evaluation). *An observable on $S = (M, \omega)$ is $A \in C^\infty(M, \mathbb{R})$. Its value in mathematical state x is the evaluation pairing*

$$\text{ev}_S : C^\infty(M, \mathbb{R}) \times M \rightarrow \mathbb{R}, \quad (A, x) \mapsto A(x).$$

The Poisson bracket is $\{A, B\}_S = \omega(X_A, X_B)$.

Lemma 2.4 (Uniqueness of Hamiltonian vector fields). *For every $H \in C^\infty(M, \mathbb{R})$ there is a unique smooth vector field X_H satisfying (1).*

Proof. At each x , nondegeneracy makes $\omega_x^\flat : T_x M \rightarrow T_x^* M$, $v \mapsto \iota_v \omega_x$, a linear isomorphism. Thus $X_H(x) = (\omega_x^\flat)^{-1}(dH_x)$ exists uniquely. In local coordinates the inverse matrix of the smooth, nonsingular matrix representing ω is smooth, so $x \mapsto X_H(x)$ is smooth. $\square$

Established physical result 2.5 (Hamiltonian evolution). *In geometric classical mechanics, Hamiltonian evolution is represented by the flow of X_H , and this flow preserves ω wherever defined [1, 4, Chs. 3–5].*

The preservation statement also has a short mathematical proof: Cartan's formula gives $\mathcal{L}_{X_H} \omega = d\iota_{X_H} \omega + \iota_{X_H} d\omega = d^2 H = 0$, hence $\frac{d}{dt}(\Phi_t^H)^* \omega = (\Phi_t^H)^* \mathcal{L}_{X_H} \omega = 0$.

3 Parallel composition

Definition 3.1 (Composite system). *For $S = (M, \omega)$ and $T = (N, \eta)$ define*

$$S \otimes T = (M \times N, \pi_M^* \omega + \pi_N^* \eta). \quad (2)$$

For kinematic processes set $(f \otimes g)(x, y) = (f(x), g(y))$. For observables define the injections

$$A \mapsto A \otimes 1 := A \circ \pi_M, \quad B \mapsto 1 \otimes B := B \circ \pi_N.$$

Here 1 is notation for independence of the other coordinate, not multiplication by a distinguished physical observable.

Lemma 3.2 (The product form is symplectic). *The two-form $\Omega = \pi_M^* \omega + \pi_N^* \eta$ in (2) is closed and nondegenerate.*

Proof. Pullback commutes with d , so $d\Omega = \pi_M^* d\omega + \pi_N^* d\eta = 0$. At (x, y) , identify $T_{(x,y)}(M \times N)$ with $T_x M \oplus T_y N$. If $\Omega((v, w), (v', w')) = 0$ for every (v', w') , choosing $w' = 0$ yields $\omega(v, v') = 0$ for every v' , hence $v = 0$. Choosing $v' = 0$ similarly gives $w = 0$. $\square$

Proposition 3.3 (Symmetric monoidal kinematics). *Hamiltonian-capable systems, smooth kinematic processes, $\otimes$, and I form a symmetric monoidal category, denoted $\mathbf{Ham}_{\text{kin}}$.*

Proof. Smooth maps contain identities and are closed under composition, giving a category. The Cartesian associator $((x, y), z) \mapsto (x, (y, z))$, unitors $(*, x) \mapsto x$ and $(x, *) \mapsto x$, and swap $(x, y) \mapsto (y, x)$ are smooth isomorphisms. Pulling product forms along these maps merely reassociates or permutes summands, so the maps are also symplectomorphisms. The pentagon, triangle, involutivity, and hexagon equations hold pointwise because both sides send each tuple to the same reordered tuple. Bifactoriality follows from $(f_2 \circ f_1) \times (g_2 \circ g_1) = (f_2 \times g_2) \circ (f_1 \times g_1)$. $\square$

Corollary 3.4 (Canonical subcategory). *The same objects and only canonical processes form a symmetric monoidal subgroupoid $\mathbf{Ham}_{\text{can}}$.*

Proof. Identities and composites of symplectomorphisms are symplectomorphisms. If $f^* \omega' = \omega$ and $g^* \eta' = \eta$, then $(f \times g)^*(\pi_{M'}^* \omega' + \pi_{N'}^* \eta') = \pi_M^* \omega + \pi_N^* \eta$. The structural maps in the preceding proof are symplectomorphisms. $\square$

4 Dynamics and factorization

Theorem 4.1 (Factorization criterion for additive Hamiltonians). *Let $H \in C^\infty(M)$ and $K \in C^\infty(N)$, and define $L = H \circ \pi_M + K \circ \pi_N$ on $M \times N$. Then*

$$X_L(x, y) = (X_H(x), X_K(y)). \quad (3)$$

Where the flows exist for the same t ,

$$\Phi_t^L = \Phi_t^H \times \Phi_t^K. \quad (4)$$

Proof. For tangent vectors (v, w) ,

$$(\iota_{(X_H, X_K)} \Omega)(v, w) = \omega(X_H, v) + \eta(X_K, w) = dH(v) + dK(w) = dL(v, w).$$

Uniqueness from the lemma gives (3). The curve $t \mapsto (\Phi_t^H(x), \Phi_t^K(y))$ has derivative (X_H, X_K) and initial value (x, y) . Uniqueness of integral curves gives (4). $\square$

New result claimed by this work 4.2 (Separation equivalence). *For a connected product $M \times N$, a Hamiltonian vector field has the split form $X_L(x, y) = (Y(x), Z(y))$ if and only if there exist $H \in C^\infty(M)$, $K \in C^\infty(N)$, and $c \in \mathbb{R}$ such that $L = H \circ \pi_M + K \circ \pi_N + c$ and $Y = X_H$, $Z = X_K$.*

Proof. The reverse implication is Theorem 4.1. Conversely, $dL = \iota_{(Y, Z)} \Omega$ gives $dL_{(x,y)}(v, w) = \omega_x(Y(x), v) + \eta_y(Z(y), w)$. Fix (x_0, y_0) and set $H(x) = L(x, y_0)$ and $K(y) = L(x_0, y) - L(x_0, y_0)$. Along any smooth path in $M \times N$, the differential of $L - H \circ \pi_M - K \circ \pi_N$ is zero by the displayed formula. Connected manifolds are path connected, so that difference is constant; its value at (x_0, y_0) is zero. Restricting dL to $(v, 0)$ and $(0, w)$ yields $\iota_Y \omega = dH$ and $\iota_Z \eta = dK$. $\square$

Proposition 4.3 (Local observables Poisson-commute). *For $A \in C^\infty(M)$ and $B \in C^\infty(N)$,*

$$\{A \circ \pi_M, B \circ \pi_N\}_{S \otimes T} = 0.$$

Proof. The defining equation and block-diagonal product form give $X_{A \circ \pi_M} = (X_A, 0)$ and $X_{B \circ \pi_N} = (0, X_B)$. Their pairing under Ω is zero. $\square$

5 Counterexamples and failure modes

Counterexample 5.1 (Interaction prevents dynamical factorization). *Take $M = N = T^*\mathbb{R}$ with coordinates (q_i, p_i) and $\omega_i = dq_i \wedge dp_i$. For*

$$L = \frac{p_1^2}{2m_1} + \frac{p_2^2}{2m_2} + \frac{k}{2}(q_1 - q_2)^2,$$

Hamilton's equations include $\dot{p}_1 = -k(q_1 - q_2)$ and $\dot{p}_2 = k(q_1 - q_2)$. When $k \neq 0$, the first component depends on q_2 and the second on q_1 . Thus the state space and symplectic form are products, while the flow is not a product flow. Composition of state spaces does not entail independent dynamics.

Counterexample 5.2 (A constraint need not be a subsystem product). *In $T^*\mathbb{R} \times T^*\mathbb{R}$, the diagonal $C = \{q_1 = q_2, p_1 = p_2\}$ is diffeomorphic to $T^*\mathbb{R}$, not to the four-dimensional ambient product. Pulling back $dq_1 \wedge dp_1 + dq_2 \wedge dp_2$ gives $2dq \wedge dp$, which is symplectic but is not the canonical form under the naive diagonal identification. Imposing a relation changes both degrees of freedom and induced geometry; it is not parallel composition.*

Counterexample 5.3 (Singular constraints leave the category). *The subset $C = \{(q, p) \in \mathbb{R}^2 : qp = 0\}$ is the union of the coordinate axes. Removing the origin produces four connected rays, whereas removing a point from a one-dimensional manifold has locally two sides. Hence C is not a smooth one-manifold at the origin. It is therefore not an object of Ham_{kin} . Treating arbitrary constraint solution sets as systems requires stratified, derived, or other enlarged geometry not supplied here.*

Counterexample 5.4 (Smooth maps do not preserve Hamiltonian structure). *The dilation $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $f(q, p) = (2q, p)$, is a smooth diffeomorphism but $f^*(dq \wedge dp) = 2dq \wedge dp$. It is a kinematic process and not a canonical process. Therefore the broad process category cannot support the claim that every process preserves Poisson brackets or Hamiltonian evolution.*

6 Empirical realization is extra structure

Definition 6.1 (Empirical realization scheme). *For a mathematical system $S = (M, \omega)$, an empirical realization scheme is a tuple*

$$\mathcal{E} = (\mathcal{P}, \mathcal{R}, \rho, \{m_A\}_{A \in \mathcal{O}}, \epsilon),$$

where $\mathcal{P}$ is a set of preparation procedures, $\mathcal{R}$ a set of records, $\rho : \mathcal{P} \rightarrow M$ a state-assignment map, $\mathcal{O} \subseteq C^\infty(M)$ a selected observable family, $m_A : \mathcal{R} \rightarrow \mathbb{R}$ readout maps, and ϵ a stated error model or tolerance. Realization means that specified predictions $A(\Phi_t^H(\rho(P)))$ agree with corresponding readouts to the criterion encoded by ϵ on a declared domain.

This is a definition of what evidence would have to connect the mathematics to data, not an assertion that a given tuple succeeds.

Empirical statement 6.2 (Domain-limited adequacy). *Finite-dimensional Hamiltonian models have experimentally successful applications to many approximately isolated classical systems, but any particular adequacy claim depends on calibration, uncertainty, regime, and comparison data. No new experimental dataset is analyzed in this paper.*

Assumption 6.3 (Operational separability). *To interpret $S \otimes T$ as two physically independent systems, assume that preparations can be varied independently, local readouts are insensitive within tolerance to remote preparation choices, and the observed joint evolution agrees within tolerance with an additive Hamiltonian on the tested domain.*

Interpretation 6.4 (Mathematical versus physical composition). *Equation (2) always defines a mathematical composite. It represents a physically realized independent composite only relative to an empirical realization scheme satisfying operational separability. Interactions may preserve the product kinematics while invalidating dynamical independence.*

Definition 6.5 (Observational equivalence relative to tests). *Given realization schemes and a chosen test family $\mathcal{T}$ of preparations, times, and observables, two models are $\mathcal{T}$ -equivalent when their predicted readout distributions or tolerance sets agree for every test in $\mathcal{T}$.*

Proposition 6.6 (Mathematical isomorphism is insufficient for physical equivalence). *A symplectomorphism $f : M \rightarrow N$ does not by itself establish physical equivalence of two realized models.*

Proof. Physical equivalence, by the preceding definition, quantifies over preparations and readouts. The datum f relates only M, ω to N, η and contains no maps between $\mathcal{P}$, $\mathcal{R}$, calibration functions, or error models. Choose identical mathematical systems and $f = \text{id}$, but assign observable q to length in one realization and to a dimensionally incompatible temperature record in another. The identity is a symplectomorphism, yet it does not preserve the typed empirical interpretation. Additional commuting relations for preparation and readout maps are necessary. $\square$

Interpretation 6.7 (Falsification conditions). *For a declared realization, reproducible violations beyond ϵ of predicted trajectories, local Poisson relations, or operational separability reject that realization on the tested domain. Such a result need not reject the abstract category: it can reject the selected Hamiltonian, observable correspondence, independence assumption, or finite-dimensional approximation.*

7 Assessment

New result claimed by this work 7.1 (Layered compositional conclusion). *Within the stated regularity assumptions, finite-dimensional Hamiltonian mechanics admits three rigorously separated layers:*

1. *symmetric monoidal kinematics, proved for products of symplectic state spaces and smooth maps;*
2. *factorized dynamics, proved exactly for additively separable Hamiltonians and characterized by the separation equivalence;*
3. *physical compositionality, which is an empirical interpretation requiring operational separability and a tolerance-bearing realization scheme.*

No implication from layer 1 to layer 2 or from layer 2 to layer 3 is asserted without the stated hypotheses.

Proof. Layer 1 is the symmetric monoidal proposition. Theorem 4.1, its converse in the separation equivalence, and the interacting-oscillator counterexample establish the boundary of layer 2. The definition of realization, the operational-separability assumption, and the insufficiency proposition establish that layer 3 contains data absent from layers 1–2. $\square$

The founding thesis is therefore mathematically coherent in this restricted case if “empirically realized” is treated as typed correspondence plus successful tests rather than as a property conferred by mathematical elegance. The present construction does not prove the thesis as a general characterization of physics. It also does not cover dissipation, stochasticity, singular reduction, gauge structure, or infinite-dimensional systems. Extending the framework to those cases is future work, not a consequence of the results above.

References

- [1] R. Abraham and J. E. Marsden, *Foundations of Mechanics*, 2nd ed., Benjamin/Cummings, 1978.
- [2] J. C. Baez and M. Stay, “Physics, topology, logic and computation: a Rosetta Stone,” in *New Structures for Physics*, Lecture Notes in Physics 813, Springer, 2011, pp. 95–174, arXiv:0903.0340.
- [3] S. Mac Lane, *Categories for the Working Mathematician*, 2nd ed., Graduate Texts in Mathematics 5, Springer, 1998, doi:10.1007/978-1-4757-4721-8.
- [4] J. E. Marsden and T. S. Ratiu, *Introduction to Mechanics and Symmetry*, 2nd ed., Texts in Applied Mathematics 17, Springer, 1999, doi:10.1007/978-0-387-21792-5.